

BANKING CUSTOMER CHURN ANALYTICS

A Data Analytics Project

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Course: Bachelor of Computer Applications (BCA)

University: Amrapali University

Domain: Banking & Financial Services

Tools Used:

Python | SQL | Power BI | Excel | Pandas | NumPy

Banking Customer Churn Analytics is a data analytics project designed to Analyze customer behavior and identify patterns related to customer churn in the banking sector. The project uses data analysis and visualization techniques to understand customer demographics, churn trends, product usage, customer segmentation, and potential financial impact. An interactive Power BI dashboard is created to present key insights and support data-driven business decisions.

- Analyze customer churn behavior.
- Identify important factors affecting customer churn.
- Analyze churn based on country, gender, and age group.
- Study customer product usage and churn patterns.
- Estimate the balance and revenue impact associated with churned customers.
- Create an interactive Power BI dashboard.
- Provide actionable insights for customer retention.

2. Dataset Description

The dataset used in this project contains customer information from a banking organization. It includes demographic, financial, and behavioral attributes of customers. The dataset is used to analyze customer churn and identify patterns that may help the bank improve customer retention.

Key attributes in the dataset include:

- Customer ID
- Age
- Gender
- Country

- Credit Score
- Balance
- Estimated Salary
- Credit Card Status
- Active Membership Status
- Number of Products
- Tenure
- Churn Status

3. Tools & Technologies

The following tools and technologies were used to perform data analysis, data processing, and visualization:

- Python: Data analysis and processing
- Pandas: Data manipulation and cleaning
- NumPy: Numerical data analysis
- SQL: Data querying and analysis
- Excel: Data inspection and basic analysis
- Power BI: Interactive dashboard creation and data visualization

4. Data Cleaning & Preprocessing

The dataset was cleaned and prepared before performing analysis. Missing values and duplicate records were checked and handled appropriately. Data types were verified, and relevant columns were selected for analysis. New calculated fields such as Age Group and key financial metrics were created to support customer churn analysis. The cleaned dataset was then used for analysis and Power BI dashboard development.

- Checked for missing values and duplicate records.
- Verified data types and column values.
- Created Age Group categories for customer segmentation.
- Prepared data for customer churn analysis.
- Created calculated measures for Churn Rate, Estimated Churned Balance, and Estimated Revenue Loss.
- Prepared the cleaned dataset for Power BI visualization.

5. Data Analysis & Methodology

The cleaned banking customer dataset was analyzed using Python, SQL, and Power BI. Exploratory data analysis was performed to understand customer demographics, churn patterns, product usage, and financial characteristics. Key metrics such as total customers, churned

customers, churn rate, churned customer balance, and estimated revenue loss were calculated. Customer segmentation was also performed based on age groups to identify different customer categories.

- Customer Demographic Analysis
- Churn Trend Analysis
- Product Usage Analysis
- Customer Segmentation
- Churn Rate Calculation
- Financial Impact Analysis
- Revenue Loss Estimation
- Executive KPI Analysis

6. Power BI Dashboard & Visualizations

An interactive dashboard was developed using Microsoft Power BI to present the results of the customer churn analysis. The dashboard provides a clear overview of customer churn patterns and their financial impact. Key Performance Indicators (KPIs) and interactive visualizations were used to make the analysis easy to understand and support business decision-making.

- **Total Customers:** Displays the total number of customers.
- **Customer Churn Rate:** Shows the percentage of customers who have churned.
- **Churned Customers:** Represents the total number of customers who left the bank.
- **Estimated Churned Balance:** Shows the estimated balance associated with churned customers.
- **Estimated Revenue Loss:** Estimates the potential financial impact of customer churn.
- **Country-wise Customer Churn:** Analyzes churn across different countries.
- **Gender-wise Customer Churn:** Compares customer churn based on gender.
- **Product Usage Analysis:** Examines the relationship between product usage and churn.
- **Age Group Analysis:** Analyzes churn patterns across different age groups.

7. Key Insights & Findings

- The overall customer churn rate is approximately **25.07%**, indicating a significant level of customer attrition.
- The dashboard shows that a considerable number of customers have churned, highlighting the need for effective customer retention strategies.
- The estimated churned customer balance indicates a significant financial value associated with customers who have left the bank.

- The estimated revenue loss demonstrates the potential financial impact of customer churn on the organization.
- Customer churn patterns vary across different countries and demographic groups.
- Product usage and customer engagement can influence the likelihood of customer churn.
- Age-group analysis helps identify customer segments that may require targeted retention strategies.
- The Power BI dashboard provides an easy-to-understand view of important KPIs and helps management make data-driven decisions.

8. Recommendations

Based on the analysis, the following recommendations can help the bank reduce customer churn and improve customer retention:

- Identify high-risk customers and provide personalized retention offers.
- Improve customer engagement through targeted communication and loyalty programs.
- Analyze customer product usage and encourage customers to use relevant banking products.
- Focus on customer segments with higher churn rates and provide customized services.
- Monitor churn trends regularly using the Power BI dashboard.
- Use customer feedback to identify service issues and improve customer satisfaction.
- Develop early-warning systems to identify customers who are likely to churn.
- Continuously track the financial impact of churn and evaluate the effectiveness of retention strategies.

9. Conclusion

- The Banking Customer Churn Analytics project successfully analyzed customer behavior and identified important patterns related to customer churn. The analysis provided insights into customer demographics, product usage, age groups, churn rate, and the financial impact of customer attrition. The Power BI dashboard presents these insights in an interactive and easy-to-understand format. The findings can help banking organizations identify high-risk customers, improve retention strategies, reduce customer churn, and minimize potential financial losses. Overall, this project demonstrates how data analytics and visualization can support effective, data-driven business decision-making.

Power BI Dashboard

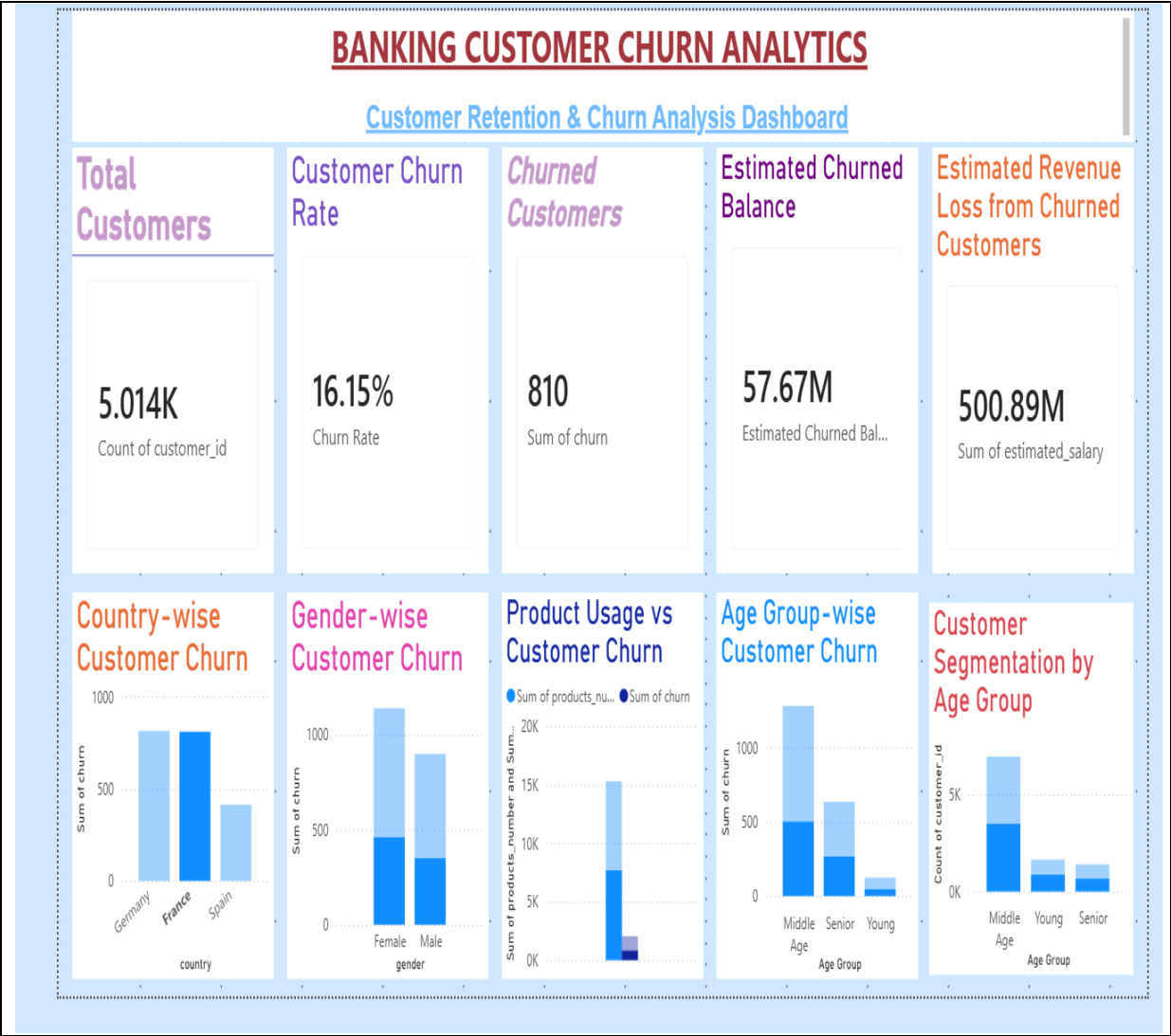


Figure 1: Banking Customer Churn Analytics Dashboard